PORTFOLIO WEBSITE
enrique-gr.github.io/portfolio

ENRIQUE GARCIA

Mechanical Engineering | Aerospace | Robotics

St. Louis, MO
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STATUS: B.S. Mechanical Engineering, Washington University in St. Louis (2027) CITIZENSHIP: U.S.

ENGINEERING EXPERIENCE

WashU Vertical Takeoff and Landing (VTOL)

Avionics & Systems Integration Lead • Vertical Flight Society DBVF | Sep 2025 -- Present

- Developing the control architecture for autonomous flight by prioritizing deterministic hardware-software interaction.
- Integrating high-rate sensor suites and custom power distribution, focusing on signal integrity and electromagnetic compatibility.
- Executing HIL simulations to validate control logic, embracing a failure-led debugging process to improve airframe stability.
- Coordinating across disciplines to ensure that mechanical design and avionics instrumentation are physically optimized.

STACK: PX4 • Python • HIL • Git • Power Systems

Multiplatform Interactive Robotics (MIRO)

Research Intern | May 2025 -- Aug 2025

- Explored multimodal signal integration by designing a 1 kHz sensor-fusion platform for EMG and IMU data.
- Derived and implemented an EKF from first principles to translate noisy raw inputs into reliable 3D motion estimates.
- Performed rigorous sensor calibration and compensation, seeking to understand the physical limits of low-cost MEMS hardware.
- Contributed to collaborative research on signal fusion presented at IEEE Body Sensor Networks (2025).

STACK: Teensy • Arduino (C++) • MATLAB • Python • Git

WashU Design Build Fly (DBF)

Aerodynamics & Payload Engineer | Sep 2024 -- Present

- Applied aerodynamics fundamentals to optimize wing geometry for high-lift mission lift/drag constraints.
- Validated structural assumptions through load analysis and FEA, iterating toward the simplest effective solution.
- Supported flight test operations to reconcile measured performance with theoretical aerodynamic predictions.
- Documented processes for the team to ensure collective technical knowledge survived beyond any single design cycle.

STACK: ANSYS • XFLR5 • MATLAB • SolidWorks • Git

TECHNICAL TOOLKIT

LOW-LEVEL / COMPUTATION
Python, C++ (20), MATLAB/Simulink, C, JS, LaTeX

PHYSICAL DESIGN & SIM
SolidWorks, ANSYS (CFD/FEA), Git, PX4, XFLR5

THEORETICAL FOUNDATIONS
Control Theory (PID/EKF), Rigid-Body Dynamics, Aerodynamics, Robotics Kinematics

SELECTED LAB WORKS

Sentinel — Target Tracking

COMPLETED

TECHNICAL EXPLORATION | 2025

A study on computer vision and predictive mathematics. Built a pipeline to detect and track fast-moving objects, using a Kalman filter to estimate 3D trajectories from noisy visual data. Frames the challenge of "intercept" through a geometry-first lens.

C++ • OpenCV • Kalman Filtering

Hyperion — 6-DOF Engine

COMPLETED

TECHNICAL EXPLORATION | 2025

An exploration of "reality in code." Architected a custom 6-DOF physics engine from scratch using quaternions and RK4 integration. Created a deterministic environment for validating GNC algorithms without reliance on commercial black-boxes.

C++20 • RK4 • Multithreading

Multi-Sensor Navigation

COMPLETED

TECHNICAL EXPLORATION | 2025

Engineered a standalone navigation unit to understand the math of "where am I?" Wrote a custom EKF to fuse inertial data, isolating the nuances of magnetometer calibration and drift compensation from high-level libraries.

C++ • Teensy • EKF • Git

WebTunnel CFD Demo

COMPLETED

TECHNICAL EXPLORATION | 2025

A browser-based tool built to make fluid dynamics intuitive. Leverages WebGL shaders to solve Navier-Stokes approximations in real-time, allowing for instant visual feedback on aerodynamic changes directly in the browser.

JavaScript • WebGL • Physics